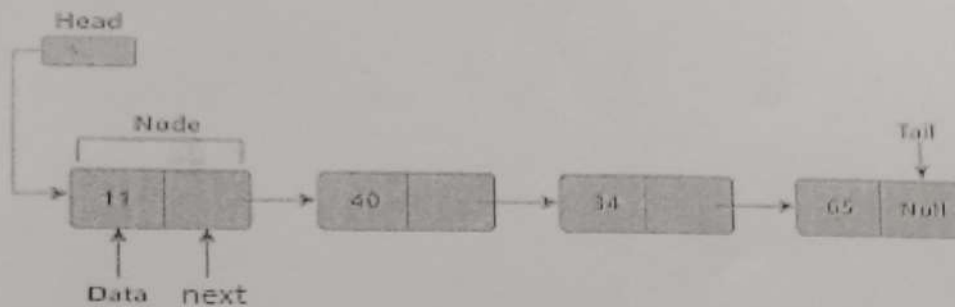


**Problem # 1:** Multiple choice question. [10 x 1pt]

- 1) In the linear queue, the value of REAR is increased by 1 when
- A. An element is deleted in a queue
 - B. An element is traversed in a queue
 - ☒ C. An element is added to a queue
 - D. An element is merged in a queue
- 2) The deletion operation in the stack is called
- A. insert
 - B. push
 - ☒ C. pop
 - D. enqueue
- 3) In a singly linked list, link fields hold pointers to the element in the linked representation of stack.
- A. Neighboring
 - ☒ B. Last
 - C. First
 - D. Middle
- 4) Which of the following is **FALSE** about Circular Queue?
- ☒ A. It helps in refilling all the free spaces by resetting REAR to zero when the REAR reaches the end of the queue
 - ☒ B. It follows the Last in First out LIFO (LIFO) principle
 - ☒ C. It connects the last position of the queue to the first position of the queue
 - ☒ D. None of the above
- 5) Which of the following is **TRUE** about linear Queues?
- ☒ A. It is also called the Last-In-Last-Out list (LILO)
 - B. It becomes full when Front = size - 1
 - C. It becomes empty only when Rear = -1 and Front = -1.
 - D. None of the above
- 6) Which of the following is **incorrect** about the stack?
- A. If top = -1, then the stack is empty
 - ☒ B. If top = -1, then the stack is not empty
 - C. pop() operation will remove (access) an element from the stack
 - D. None of the above



- 7) Use the following LinkList to determine the value of following instructions
 node <I> *q;
 q= head-> next->next;
 cout<<q->next->data;



- A. 65
 B. 40
 C. 34
 D. 11
- 8) Undo/Redo operations in Excel or Word processing programs are a real example of linear queue data structure.
 A) True
 B) False
- 9) Linked lists allow random memory access.
 A) True
 B. False
- 10) Arrays are not cache friendly.
 A. True
 B) False

7

*** Write the correct answers to the problem # 1 in the table below:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
c	c	b	b	A	b	A	A	A	b



Problem # 2 If the C++ program below represents the implementation of the circular queue, write the output of this code. [5pts]

```
#include <iostream>
using namespace std;
const int size = 6;
template<class T>
class CircularQ
{
private:
    T q[size];
    int front, rear;
    int count;
public:
    CircularQ()
    {
        front = 0;
        rear = size - 1;
        count = 0;
    }
    bool isFull()
    {
        return (count == size);
    }

    bool isEmpty()
    {
        return (count == 0);
    }
    void enqueue(T item)
    {
        if (isFull())
        {
            cout << "full\n";
        }
        else
        {
            rear = (rear + 1) % size;
            q[rear] = item;
            count++;
        }
    }
    int dequeue()
    {
        if (isEmpty())
        {
            cout << "empty";
            return 0;
        }
        else
        {
            int x = q[front];
            front = (front + 1) % size;
```



```

count--;
return x;
}
int getSize()
{
return count;
} void print_queue()
{
for (int i = front; i <= rear; i++)
cout << "Position : " << i << " , Value : " << q[i]<<endl;
}
};

```

```

int main()
{
CircularQ <int> qq;
qq.enqueue(72);
qq.enqueue(45);
qq.enqueue(17);
qq.enqueue(11);
qq.enqueue(2);
qq.enqueue(31);
qq.enqueue(9);
qq.enqueue(81);
for(int i=0; i<=qq.getSize()/2; i+=2)
cout << qq.dequeue() << endl;
qq.print_queue();
cout<<qq.dequeue()<<"\n";
qq.enqueue(32);
qq.enqueue(35);
cout<<qq.dequeue();
return 0;
}

```

31 2 11 17 45 72

Output

Handwritten output and calculations:

17
11
2
31
~~31 2 11 17 45 72~~
35 38 45 72

Handwritten calculations:

2.5
5

**Problem # 3:** Write the output of the below C++ code. [5pts]

```
#include <iostream>
using namespace std;
template <class T, class U, class V = char>
class ClassTemplate {
private:
    T var1;
    U var2;
    V var3;
public:
    ClassTemplate(T v1, U v2, V v3)
    {
        var1=v1;
        var2=v2;
        var3=v3;
    }
    void printVar() {
        cout << "var1 = " << var1 << endl;
        cout << "var2 = " << var2 << endl;
        cout << "var3 = " << var3 << endl;
    }
};
int main() {

    ClassTemplate<int, double> obj1(20, 19.8, 'd');
    cout << "obj1 values: " << endl;
    obj1.printVar();
    ClassTemplate<double, char, bool>
    obj2(12.8, 'c', true); cout << "\nobj2 values: " << endl;
    obj2.printVar();
    return 0;
}
```

Output

obj1 values:

var1 = 20

var2 = 19.8

var3 = 'd'

obj2 values:

var1 = 12.8

var2 = 'c'

var3 = true

5



Problem # 4: Write a C++ program to find the maximum element stored in the stack. [5pts]

```
include <iostream>
using namespace std;
```

```
int main() {
    int y;
```

```
for (i=0; i<tos; i++)
```

```
pop();
y = tos;
pop();
```

```
if (IsEmpty())
```

```
cout << "maximum num" << y << endl;
```

Problem # 5: Write a C++ program to find the average of all elements stored in the queue? [5pts]

```
front = -1;
rear = -1;
```

```
for (i=0; front != rear; i++)
```

```
dequeue();
sum = sum + dequeue();
count = count + 1;
```

```
average = sum / count
```

```
cout << "average of all elements" << average << endl;
```

~~count the distance between~~
sum all the element between front and rear
and divide by the num of elements

for (i=front+1; i<rear; i++)

14/5

Good Luck